ROC-AUC Curve

Why ROC Curve?

Threshold selection - It is related to classification

* True Positive Rate (TPR) :- Benefit

Predicted

Actual	1	0
1	TP	FN
0	FP	TN

$$\boxed{TPR = \frac{TP}{TP+FN}}$$

↑ Good thing

eg.
Out of 100 spams how many actually model predicted correctly.

False Positive Rate (FPR): Cost

Predicted

Actual	1	0	
	TP	FN	1
	FP	TN	0

$$\boxed{FPR = \frac{FP}{FP + TN}}$$ ↓ Good thing

eg
Out of 100 non spams, how many

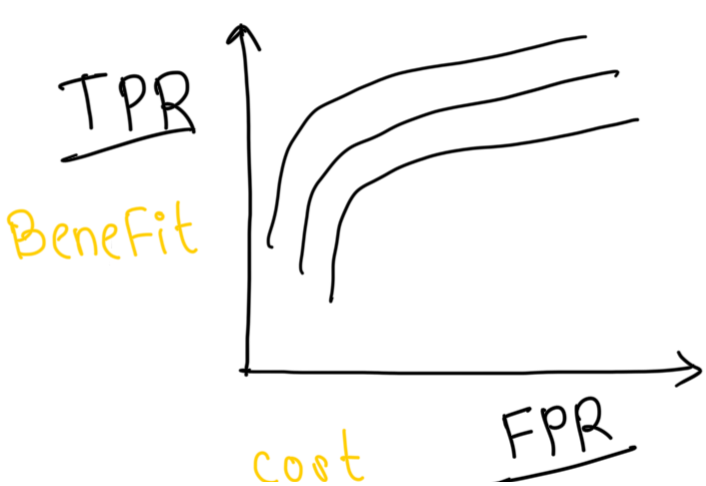
got predicted as spam by model

TPR → Benefit ↑

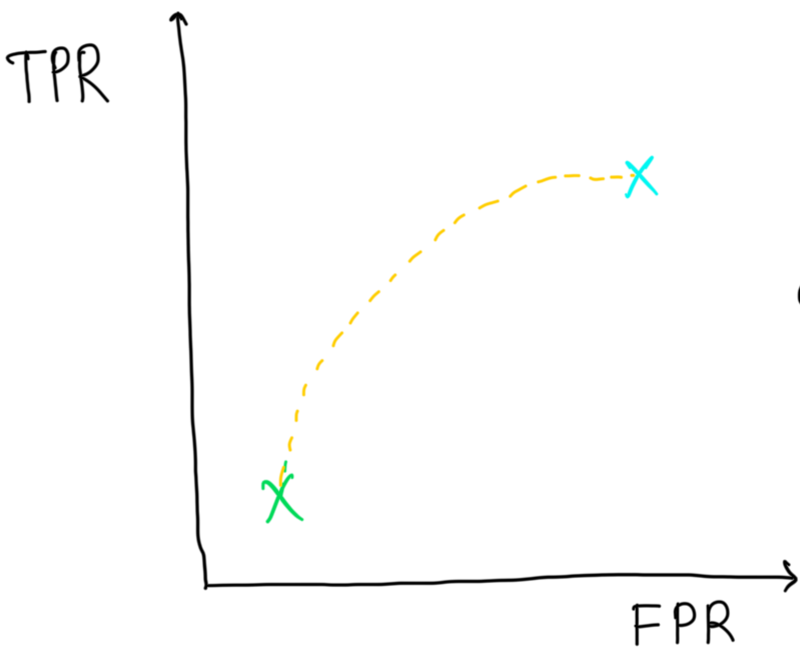
FPR → Cost ↓

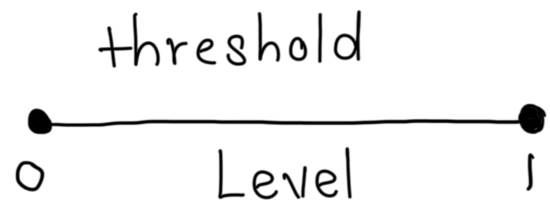
ROC Curve C Receiver Operator Characteristics)

Graph plotted between TPR and FPR is called ROC Curve



* Different Cases





email spam

0.1 → Threshold

0.2 → Spam

0.09 → Not spam

$TP \uparrow \quad \therefore \quad TPR = \dfrac{TP}{TP + FN}$ Increase

$$FPR = \frac{FP}{FP + TN}$$ Increase

Decrease threshold → FPR ↑
TPR

Increase threshold → FPR ↓
TPR

AUR-ROC

It measures the entire 2D area underneath the entire ROC curve from (0,0) to (1,1). AUC provides an aggregate measure of performance across all possible classification thresholds.

- An AUC of 1 indicates that the model

has perfect discrimination: It correctly classifies all +ve & −ve instances.

- An AUC of 0.5 suggests that the model has no discrimination ability: It is as good as random guessing

- An AUC of 0 indicates that the model is perfectly wrong: It classifies all positive instances as negative and all negative instances as +ve.

In practice, AUC values usually fall bet 0.5 (random) & 1 (perfect), with higher values indicating better classification.